

ArcWatch AI

NETWORK READINESS INTELLIGENCE

The grid is dynamic. Risk is not.

Legacy operations struggle to synthesize high-velocity data into actionable response.



Data Fragmentation

Sensor data, weather feeds, and asset logs live in silos, preventing a unified operational picture.

60%

Time spent searching for data

Reactive Maintenance

Response is triggered by failure rather than risk, leading to costly emergency repairs and downtime.

4.2x

Cost of reactive vs. planned work



Climate Exposure

Increasing frequency of extreme weather events creates unpredictable stress on aging infrastructure.

30%

Increase in weather-related outages

One command center for readiness



Validated Asset Data

Real-time monitoring of 15+ critical grid zones with automated sensor validation.

Live Risk Analytics

Composite risk scoring based on temperature, vibration, and weather exposure.

Readiness Distribution

Instant visibility into asset posture from Critical to Operational.

Response Planning

Direct link between risk detection and maintenance crew positioning.

From validated data to action



Key Feature: Selected Asset Snapshot

T-104

Central Substation

CRITICAL

Temperature

96°C

Vibration

8.1 mm/s

Weather Risk

High

Composite Score

100/100

RECOMMENDED ACTION

Inspect within 6 hours and pre-position a maintenance crew.

Plan Response

IBM Technologies



Hybrid Cloud

- IBM Cloud Satellite for edge data ingestion
- Red Hat OpenShift for scalable microservices
- Secure cross-zone data synchronization



watsonx.ai

- Predictive risk modeling for asset failure
- Natural language insights for operators
- Automated anomaly detection in sensor streams



watsonx.data

- Unified lakehouse for historical asset logs
- High-performance querying for live analytics
- Governance and lineage for validated data

Results / Impact

15

Assets Monitored

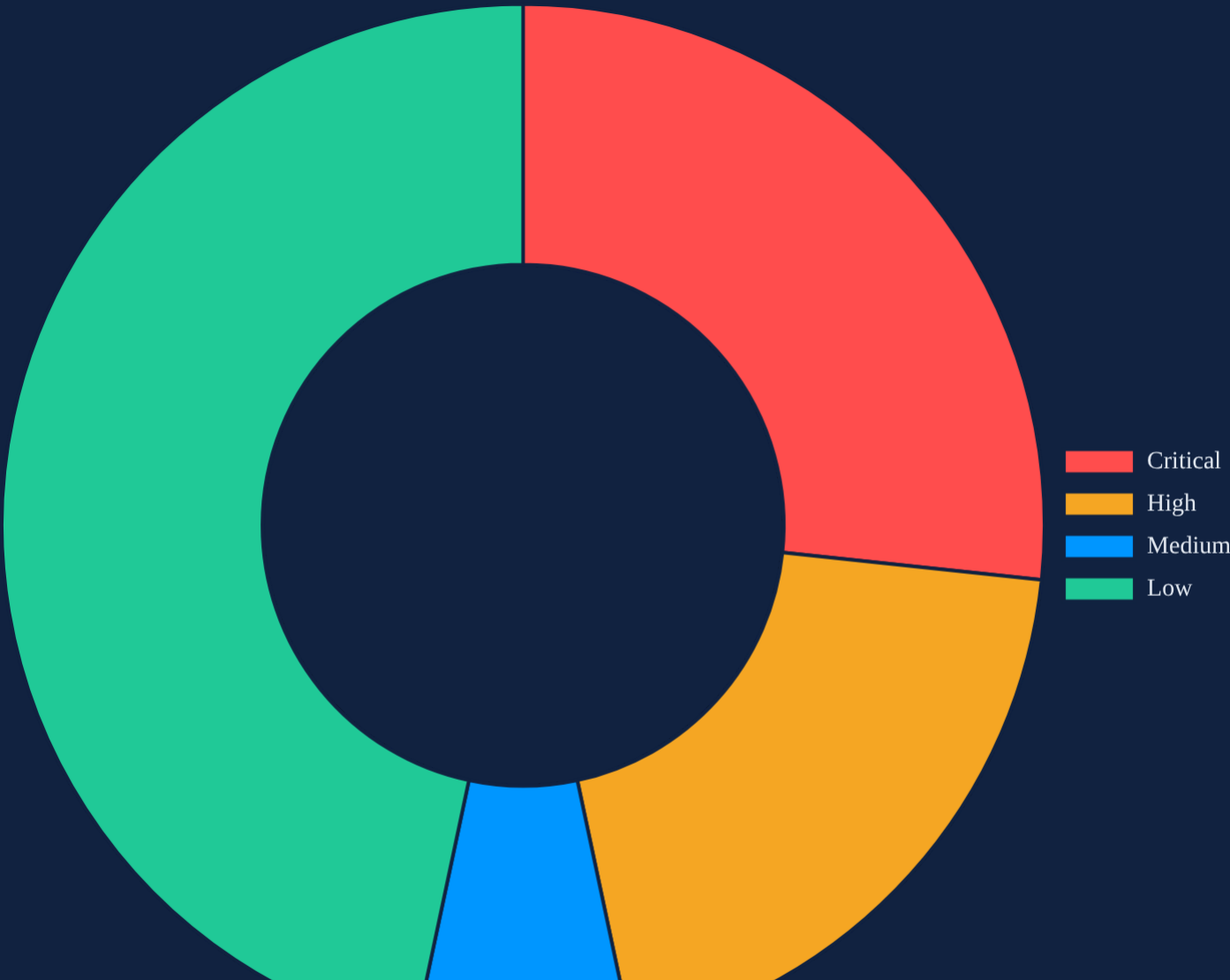
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Critical Assets Identified

6

High Weather Exposure Zones

Readiness Distribution



NEXUS FOUR



Het Patel

LEAD ENGINEER

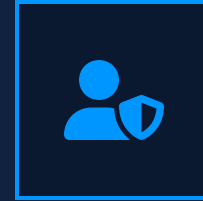
Built the core Risk Engine and sensor validation pipeline.



Dweeja Shingala

PRODUCT DESIGNER

Designed the Command Center UI and asset snapshot flow.



Birva Vachhani

DATA SCIENTIST

Developed the predictive failure models using watsonx.ai.



Yug Vasani

OPS SPECIALIST

Architected the crew response and weather exposure logic.